HELPMATE MOVIE ASSIST





Project Report

By

SAI MITRA JANDHYALA

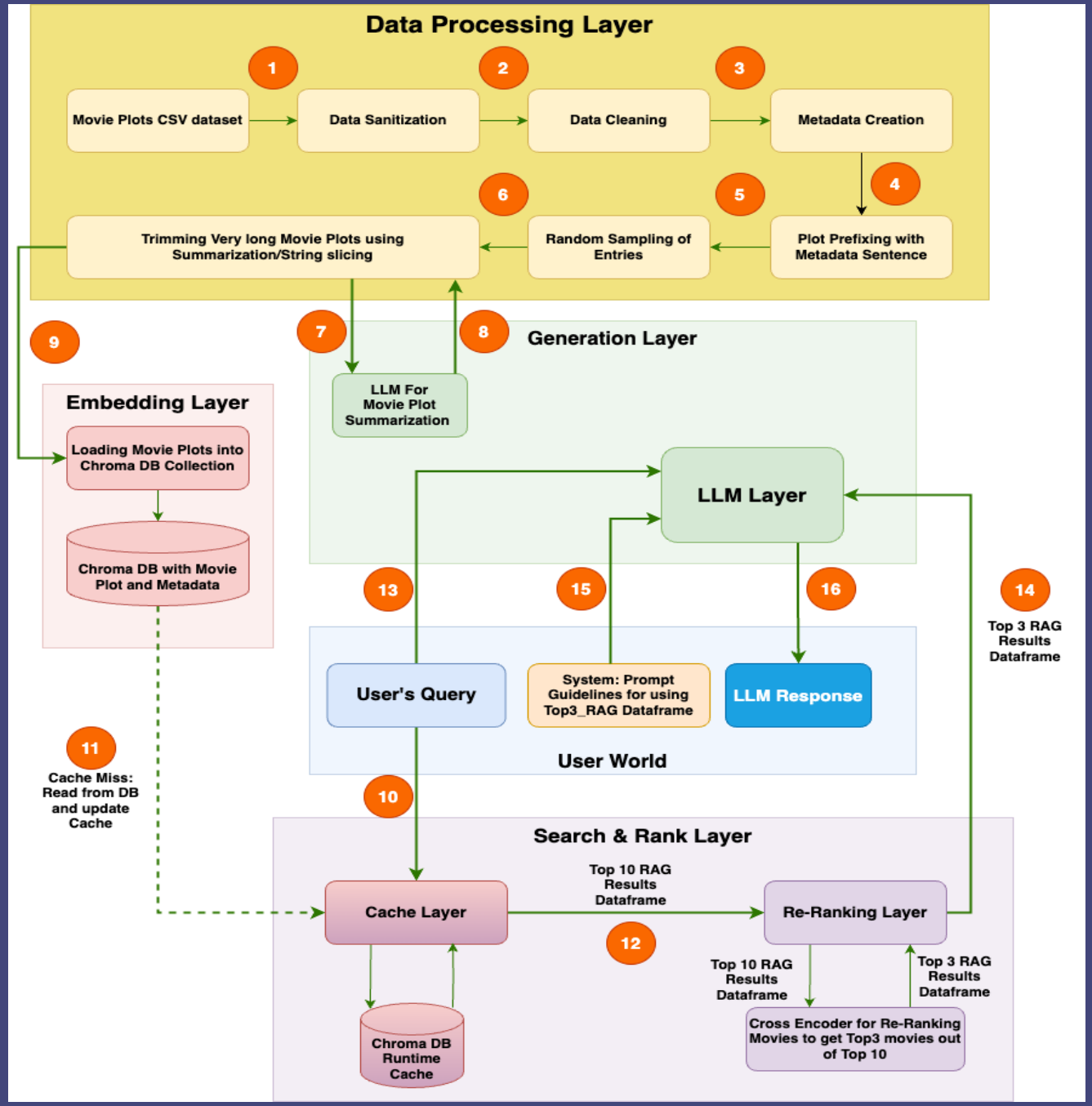
HelpMate (Option 1: BYOP)

PROJECT OVERVIEW

Chatbot based HelpMate Application for querying Movies based on Plots

- The goal of the project is the provide RAG based search on Movie database containing movie plots. User's query would represent the kind of movies the user likes to watch, and this would be semantically compared with the existing Movie plots in the vector database.
- Movie watchers surf movies based on their current mood, and they might want to have a tool which can take in their query about the type of movies that interest them and present them with some options.
- Movie plots in general contain the overall content of the Movie and user's queries can be matched against the plot to narrow down the best choices in a set of movies.
- Movie plots can be long and need to trimmed down through text summarization capabilities of LLMs or can be trimmed down with string operations.
- Since user's queries are in English, semantic search techniques make more powerful queries that are based on the meaning of the user's query rather than the words in the query.
- Semantic Search combined with LLM generated response make a good combo for a chat based tool which can suggest movies based on plots.

PROJECT DESIGN



DESIGN EXPLANATION

HelpMate Movie Assist Design

- Steps 1 to 9 in the design are Data Processing Layer where the Movie data obtained from Kaggle Movies Plot dataset https://www.kaggle.com/datasets/jrobischon/wikipedia-movie-plots
  - Lot of steps to sanitize the data, impute missing values with “unknown”
  - Metadata addition which composes all the details of the movie like year, genre, ethnicity, cast which will be used in Chroma DB creation.
  - Removing movies with plots of word length < 5
  - Trimming movie plot with word length > 10
    - Text Summarization was used but there are limitations on RPM on OpenAI Summarization feature.
    - Text trimming using word count cap of 1000 words was instead used to simplify the plot size reduction.
  - Prefixing movie metadata in sentence form to the plot helped in creating better results. LLM was inefficient to investigate “Metadata” column of dataframe to infer “Title” or “Genre” or other movie details.
  - Sampling the data as the dataset as 30k+ movies and OpenAI ChromaDB embedding function has RPM limitations.
  - Once the data is clean, the movie plots and their metadata are
  - Steps 10 to 16 start from taking user’s query as input followed by checking in Cache if a similar query exists and if not, collecting top 10 movie results from ChromaDB and updating the cache. Following the query results are re-ranked using Cross-Encoder to get top3 movies dataframe.
  - Top 3 movies along with user query and system prompt guidelines are fed into an LLM which then processes the user’s query and Top 3 dataframe and produces LLM Response that adheres to the guidelines placed by the system prompt.

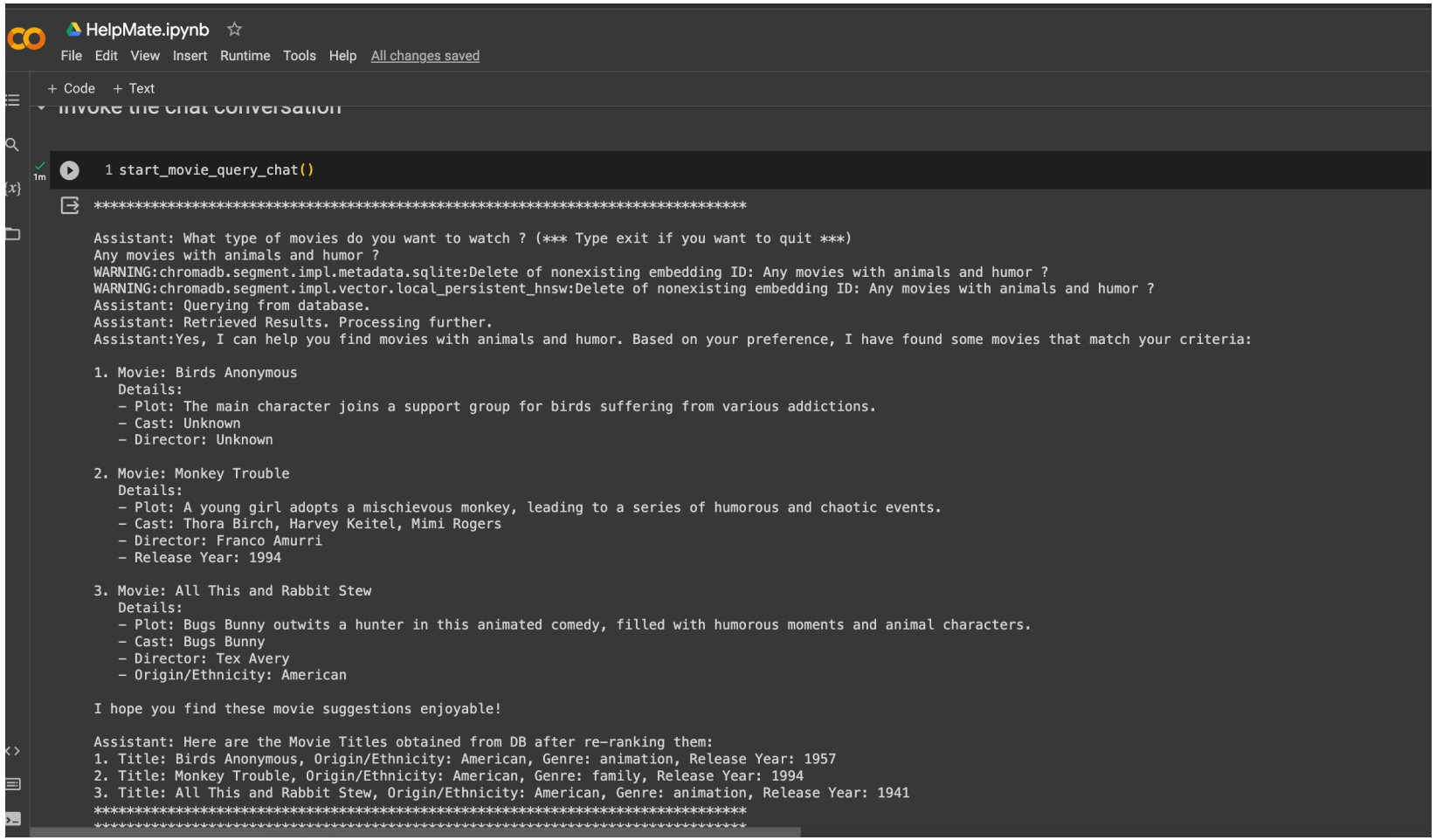
DESIGN EXPLANATION

Challenges & Lessons Learnt

- Data Cleaning: Datasets are prone to having inconsistent data. Lot of cleaning needs to be applied before using it as a feed for chatbot.
- OpenAI responses are not always constant, and they vary. Hence it important to set prompt guidelines so that LLM does not deviate from its goal of presenting a set of movies based on user's query.
- OpenAI conversation has a limit of tokens and care should be taken to complete the goal within those token limits. Hence 30K+ movie entries cannot all be processed by the embedding function used in creation ChromaDB. Only a small sample of 1500-1800 entries was found to be working.
- Likewise, OpenAI Embedding function also has usage limits which need to be considered.
- OpenAI Text summarization has RPM limitations, hence a simpler plot trimming method was used to trim words beyond 1000 for movies with more than 1000 words.
- LLMs was not able to identity movie plots and movie details correctly. Hence Movie metadata was converted as a sentence and pre-fixed to movie plot which then gave proper results as LLM was able to infer the movie details within the plot itself.

DESIGN EXPLANATION

Sample Chat Response for User's query "Any movies with animals and humor ?"



THANK YOU !

